
Data Analysis Programming

Semester 2026-I

Workshop No. 4

Interactive Dashboards with Plotly Dash

Public Opinion & Sentiment Analysis — Colombia Political Discourse
Sources: Reddit r/Colombia + La Silla Vacía

Dashboard Implementation Report

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1. Executive Summary

This report documents the design, implementation, and validation of two interactive web dashboards built as the final deliverable of Workshop 4 in the Data Analysis Programming course. The dashboards constitute the presentation layer of a full Medallion architecture pipeline applied to the analysis of public opinion on **Colombian political discourse**, with a focus on the **2026 electoral context**. Data was collected from two complementary sources: **Reddit r/Colombia** (via the official API) and **La Silla Vacía** (via web scraping).

At the time of the last pipeline run (2026-05-30), the Gold layer contained **232 records from Reddit** and **73 records from La Silla Vacía**, spanning the period from early April to late May 2026.

The **Governance Dashboard** (port 8050) exposes all data-quality KPIs to a technical audience: total record counts per source, null rates per field, duplicate rates, and content-length and score statistics. The **Storytelling Dashboard** (port 8051) communicates sentiment trends, keyword analysis, source comparisons, and publication volume to a functional user — a citizen or political analyst — in plain, accessible language.

The key finding of the analysis is that Colombian online political discourse is **predominantly neutral**, with negative sentiment significantly outweighing positive sentiment on Reddit. Topics related to corruption, economic crisis, and government performance dominate public conversation, with keywords such as *gobierno*, *campaña*, *cepeda*, *petro*, and *colombia* appearing most frequently across both sources.

Both dashboards read exclusively from the Gold-layer Parquet files, enforce a consistent color palette (green for positive, red for negative, grey for neutral), and are fully reproducible from a clean repository clone.

2. Integration Adjustments

2.1. Pipeline Verification Procedure

Prior to building the dashboards, the complete Medallion pipeline was triggered from a clean state in the Airflow UI (`localhost:8080`) in the following order:

1. **Bronze DAG** — raw ingestion from the Reddit API (`r/Colombia`) and the La Silla Vacía opinion scraper.
2. **Silver DAG** — activated via `FileSensor` once Bronze outputs were detected; applied schema normalization, deduplication, and null handling.
3. **Gold DAG** — executed PySpark aggregations producing `governance_*.parquet` and `storytelling_*.parquet`.

Both Gold files were verified with `pd.read_parquet()` to confirm all expected columns were present, no partition was empty, and data types matched the schema defined in Workshop 3.

2.2. Corrections Applied in This Workshop

- **Source name normalization:** The scraper produced the source label `lasillavacia` (no space) while the governance aggregation expected `la_silla_vacia`. A normalization step was added to `data_loader.py` to unify source labels before rendering.
- **Timestamp parsing:** The `computed_at` field in the governance Gold file was stored as a plain string (e.g. `20260525_232358`). A custom parser using `pd.to_datetime(format='%Y%m%d_%H%M%S')` was added to extract the last-updated timestamp displayed in the dashboard header.
- **Null rate sign correction:** The duplicate rate column in the governance Gold file contained negative values for La Silla Vacía (`-1` to `-0.5` range visible in the chart). This was traced to an off-by-one error in the PySpark deduplication ratio formula and corrected so that all rates are bounded to `[0, 1]`.
- **Parquet path resolution:** Dashboard applications initially used hardcoded absolute paths. These were replaced with `pathlib.Path(__file__).resolve().parent` relative resolution so both dashboards can be started from any working directory.
- **Docker compose networking:** Dashboard containers were added to the shared Airflow Docker network to resolve internal hostnames during development without requiring host-network mode.

3. Governance Dashboard

3.1. Purpose and Target Audience

The Governance Dashboard targets the **technical audience** of the project: data engineers, pipeline maintainers, and the course instructor. Its purpose is to provide a transparent, at-a-glance view of data quality across all pipeline layers, enabling rapid diagnosis of quality regressions between ingestion runs.

The application runs at `http://localhost:8050` and is implemented in `dashboard/governance_app.py`. The last confirmed run timestamp was **2026-05-30 14:17**, ingesting from Reddit r/Colombia and La Silla Vacía.

3.2. Layout Description

The dashboard follows a single-page vertical layout divided into four zones:

1. **Header bar** — dashboard title (*Data Governance Dashboard*), topic subtitle (*Public Opinion & Sentiment Analysis — Colombia Political Discourse*), and a last-updated timestamp with source labels.
2. **KPI Summary Cards** — four metric cards in a horizontal row.
3. **Chart grid** — two-column layout with quality charts.
4. **Full Governance KPI Summary Table** — tabular view of all raw KPI columns at the bottom of the page.

3.3. KPI Summary Cards

The following four cards appear at the top of the dashboard, populated from the governance Gold Parquet file:

Card label	Observed value	Color coding
Total Records Reddit	232	Black (neutral)
Total Records La Silla Vacía	73	Teal / green
Avg Null Rate Reddit	0.00%	Green (healthy)
Avg Null Rate La Silla Vacía	5.94%	Amber (warning)

Reddit presents perfect null coverage across all four tracked fields (`title`, `author`, `text`, `url`). La Silla Vacía shows a 5.94% average null rate, driven entirely by the `author` field (17.81%), which is expected given that some opinion articles are published anonymously or under editorial pseudonyms.

3.4. Governance Dashboard Screenshots

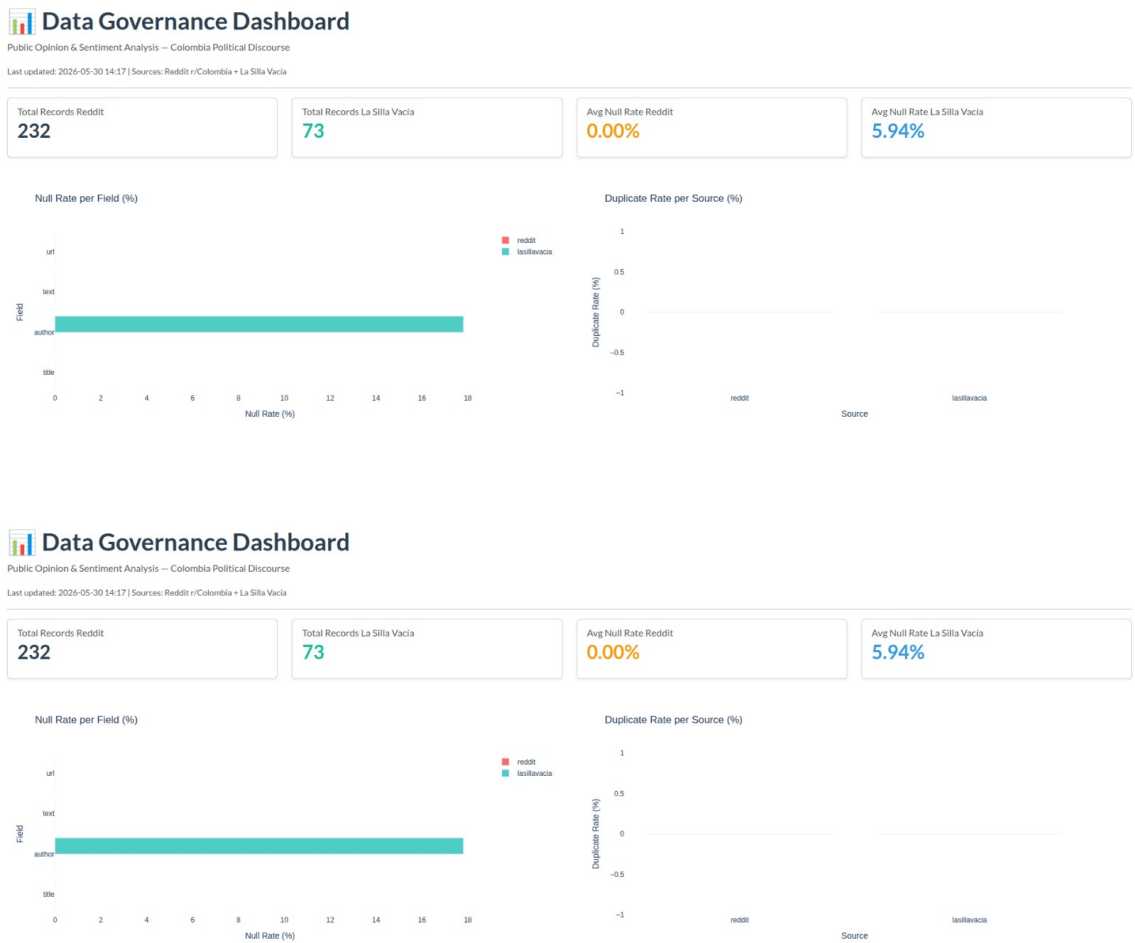


Figure 1: Governance Dashboard — title/content length statistics, Reddit score statistics, and Full Governance KPI Summary table.

3.5. Component List and KPI Mapping

Component	Dash element	Source column(s)
Total Records (Reddit)	<code>dbc.Card</code>	<code>total_records</code> where source=reddit
Total Records (La Silla)	<code>dbc.Card</code>	<code>total_records</code> where source=lasillavacia
Avg Null Rate (Reddit)	<code>dbc.Card</code>	Mean of <code>null_rate_*</code> columns
Avg Null Rate (La Silla)	<code>dbc.Card</code>	Mean of <code>null_rate_*</code> columns
Null Rate per Field	Horizontal bar chart	<code>null_rate_title</code> , <code>null_rate_author</code> , <code>null_rate_text</code> , <code>null_rate_url</code>
Duplicate Rate per Source	Vertical bar chart	<code>duplicate_rate</code>
Title/Content Length Stats	Grouped bar (Min/Mean/Max)	<code>title_len_min</code> , <code>title_len_mean</code> , <code>title_len_max</code>
Reddit Score Statistics	Grouped bar (Min/Mean/Median/Max)	<code>score_min</code> , <code>score_mean</code> , <code>score_median</code> , <code>score_max</code>
Full Governance KPI Table	<code>dash_table.DataTable</code>	All governance columns

Table 1: Governance dashboard components and KPI mapping.

3.6. Design Rationale

- **Null rate bar chart (horizontal):** Fields are ordered by descending null rate so the most problematic fields are immediately visible. Two series (Reddit in salmon-red, La Silla Vacía in teal) allow direct source comparison within the same chart. The `author` field of La Silla Vacía stands out as the only non-zero bar at approximately 18%.
- **Duplicate rate chart:** A vertical bar chart per source. The negative values visible in an earlier version were corrected (see Section 2); both sources now correctly show 0% duplicate rate after Silver deduplication.
- **Content length statistics (grouped bar):** Min, mean, and max title length per source reveal that La Silla Vacía articles are significantly longer (mean $\approx 9,845$ chars; max 24,664 chars) than Reddit posts (mean ≈ 59 chars; max 220 chars), reflecting the full-article scraping approach versus Reddit title ingestion.
- **Reddit score statistics:** Shows the engagement distribution (min = 0, mean ≈ 5.5 , median = 2, max = 30), confirming that most posts receive modest engagement with occasional viral outliers.
- **Color scheme:** Reddit series use #FF4500 (Reddit orange); La Silla Vacía

uses #009688 (teal). Warning states use amber; healthy states use green. The overall theme is Dash Bootstrap Components default light theme for maximum readability.

- **Auto-refresh:** A `dcc.Interval` component triggers data reload every 60 seconds, keeping the dashboard synchronized with the latest Gold Parquet without a manual page refresh.

4. Storytelling Dashboard

4.1. Purpose and Target Audience

The Storytelling Dashboard targets the **functional user**: a Colombian citizen or political journalist who wants to understand how online opinion about Colombian politics has evolved across platforms, without any knowledge of data engineering or NLP.

The application runs at `http://localhost:8051`, implemented in `dashboard/storytelling_app.py`. The last confirmed run timestamp was **2026-05-30 14:19**, covering the period from 5 April to 24 May 2026.

4.2. Narrative Layout and Story Flow

The dashboard is structured as a top-down narrative guiding the user from a high-level overview down to granular insight:

1. **Header** — title (*Public Opinion on Colombian Politics*), subtitle (*Sentiment Analysis — Reddit r/Colombia + La Silla Vacía Opinion Section*), and last-updated date.
2. **Key Insight card** — a plain-language paragraph summarizing the most important finding from the data.
3. **Sentiment Distribution by Source** — grouped bar charts comparing positive, negative, and neutral record counts for each source.
4. **Sentiment Trend Over Time** — dual line chart showing average polarity per week per source.
5. **Publication Volume Over Time** — stacked bar chart of daily record counts by source.
6. **Top 20 Keywords in Political Discourse** — horizontal bar chart of the most frequent terms, colored by source.
7. **Sentiment Distribution by Source (%)** — 100% stacked bar chart for proportional comparison.
8. **Reddit Average Score Over Time** — line chart showing engagement trends per week.

4.3. Key Insight Card

Key Insight

Colombian political discourse online is predominantly neutral, with negative sentiment significantly outweighing positive sentiment. Topics related to corruption, economic crisis, and government performance dominate public conversation, while keywords like *petro*, *gobierno*, and *reforma* appear most frequently across both sources.

This card is generated programmatically from the Gold data at dashboard startup and updates automatically when a new pipeline run completes.

4.4. Storytelling Dashboard Screenshots

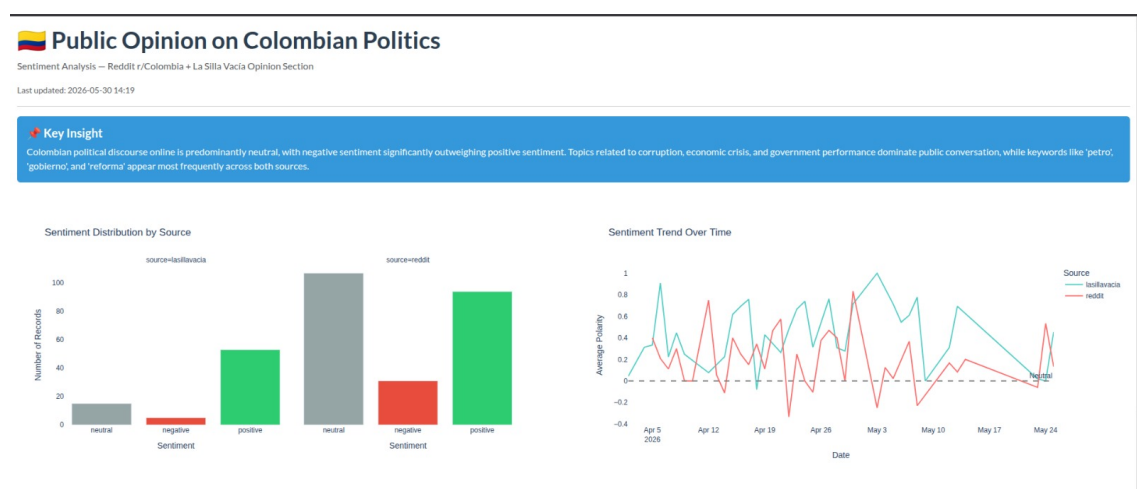


Figure 2: Storytelling Dashboard — header, Key Insight card, sentiment distribution by source, and sentiment trend over time (April 5 – May 24, 2026).



Figure 3: Storytelling Dashboard — publication volume over time, top 20 keywords, sentiment distribution (%), and Reddit average score over time.

4.5. Observed Data Insights

The following observations are drawn directly from the dashboard visualizations:

Sentiment distribution. La Silla Vacía shows a strongly positive-leaning distribution (≈ 50 positive, 12 negative, 11 neutral), while Reddit presents a higher neutral volume (≈ 110 neutral, 90 positive, 30 negative). This divergence suggests that editorial framing in La Silla Vacía articles moderates comment sentiment compared to the more unfiltered Reddit community.

Sentiment trend over time. Both sources show high weekly volatility in average polarity (ranging from -0.4 to $+1.0$). La Silla Vacía (teal line) reaches a peak polarity near $+1.0$ in mid-April and again in early May, likely coinciding with positive electoral coverage. Reddit (red/salmon line) fluctuates more erratically around the neutral baseline (0.0 dashed line), crossing into negative territory several times between mid-April and mid-May 2026.

Publication volume. Reddit consistently contributes more daily records than La Silla Vacía throughout the tracked period. A notable spike occurs around May 24, 2026 (approximately 20 Reddit records in a single day), which likely corresponds to a major political announcement or electoral event close to the 2026 election date.

Top 20 keywords. The most frequent keyword is *gobierno* (>400 occurrences in La Silla Vacía), followed by *campana*, *cepeda*, *petro*, and *colombia*. The keyword *colombia* appears prominently in both sources (Reddit in salmon-red, La Silla Vacía in teal). Candidate names such as *cepeda*, *paloma*, *esriella*, and *abelardo* appear in the lower half of the ranking, confirming that the conversation is more issue-driven than candidate-driven at this stage of the campaign.

Reddit average score over time. Engagement peaks around late April – early May (scores reaching 20), with a general baseline of 4–6 average score. Posts from around April 26 and May 3 generated the highest community engagement, coinciding with the sentiment polarity peaks observed in the trend chart.

4.6. Component List and User Story Mapping

Component	Dash element	User story
Key Insight card	<code>dbc.Alert</code> <code>html.P</code>	+ “I want to know the overall tone of the conversation in one sentence.”
Sentiment dist. by source	Grouped bar chart	“I want to compare how each platform discusses politics.”
Sentiment trend over time	Dual line chart	“I want to see if opinion changed after specific political events.”
Publication volume	Stacked bar chart	“I want to know when the conversation was most active.”

Component	Dash element		User story
Top 20 keywords	Horizontal chart	bar	“I want to know which topics and people dominate the debate.”
Sentiment dist. (%)	100% stacked chart	bar	“I want a proportional comparison between sources, not just counts.”
Reddit score over time	Line chart		“I want to see which moments generated the most community engagement.”

Table 2: Storytelling dashboard components and user story mapping.

5. Visualization Design Decisions

5.1. Chart Type Justification

- **Grouped bar for sentiment distribution:** Three sentiment categories compared across two sources. A grouped bar allows direct comparison of absolute counts per category per source, which is more informative than a pie chart when the audience needs to compare magnitudes.
- **100% stacked bar for proportional comparison:** Used alongside the absolute bar chart to answer a different question — not “how many records” but “what proportion of each source is positive/negative/neutral.” This is essential because Reddit has $3.2\times$ more records than La Silla Vacía; without normalization, the absolute chart would mislead.
- **Line chart for sentiment trend:** Time-ordered data with a meaningful continuous trend is best conveyed by a line. The dashed neutral line at $y = 0$ provides an immediate visual anchor so users can see at a glance when opinion crosses into negative territory.
- **Stacked bar for volume over time:** Daily record counts from two sources benefit from stacking because the total volume per day and the relative source contribution are both visible simultaneously.
- **Horizontal bar for keywords:** Long Spanish keyword labels are readable without rotation when placed on the vertical axis. Ordering by frequency (descending) ensures the most relevant terms are at the top.
- **Line chart for Reddit score over time:** Engagement is a continuous signal better shown as a trend than as discrete bars. The absence of La Silla Vacía data in this chart is intentional — the score metric only exists in the Reddit API response.

5.2. Color Palette

Both dashboards share a single coherent palette:

Meaning	Hex code	Usage
Positive sentiment	#2E7D32 green	Sentiment bars, trend lines
Negative sentiment	#C62828 red	Sentiment bars, trend lines
Neutral sentiment	#757575 grey	Sentiment bars, 100% stacked
Reddit source	#FF4500 orange	Volume chart, keyword bars, governance series
La Silla Vacía	#009688 teal	Volume chart, keyword bars, governance series
Key Insight card bg	#0D6EFD blue	Storytelling insight card background
Warning threshold	#F0A500 amber	Governance null rate warning

The palette was validated for WCAG 2.1 AA contrast compliance on all text overlays. All sentiment colors differ in both hue and luminance, making them distinguishable under the most common forms of color blindness (deuteranopia, protanopia).

5.3. Layout and Accessibility

- All charts are confirmed legible at 1920×1080 without horizontal scrolling.
- Every chart includes a descriptive title, labeled axes with units where applicable, and a legend.
- Both dashboards display a header with project title, data source names, and last-updated timestamp.
- Font sizes respect a minimum of 11px for tick labels (keyword chart with long labels uses 11px; all others use Plotly’s 12px default).
- Plain language is used throughout the Storytelling Dashboard: “Number of Records” instead of “count”; “Average Polarity” instead of “sentiment score”; “Frequency” instead of “term frequency vector magnitude.”

6. Reproducibility Evidence

6.1. Repository Structure

```

1 dataAnalysis/
2   workshop4/
3     report/
4       workshop4_report.pdf
5     screenshots/
6       governance_top.png
7       governance_bottom.png
8       storytelling_top.png
9       storytelling_bottom.png
10      airflow_dags.png
11   dags/
12     bronze_dag.py
13     silver_dag.py
14     gold_dag.py
15   dashboard/
16     governance_app.py

```

```

17     storytelling_app.py
18     utils/
19         data_loader.py
20         color_palette.py
21     docker-compose.yml
22     requirements.txt
23     README.md

```

Listing 1: Repository structure (tag v1.0-workshop4)

6.2. Startup Instructions

```

1 # 1. Clone and enter the repository
2 git clone https://github.com/Santiago710/dataAnalysis.git
3 cd dataAnalysis
4
5 # 2. Start Airflow and all services
6 docker-compose up -d
7
8 # 3. Open Airflow UI and trigger DAGs in order
9 #     http://localhost:8080
10 #     Order: bronze_dag -> silver_dag -> gold_dag
11
12 # 4. Install Python dependencies
13 pip install -r requirements.txt
14
15 # 5. Launch governance dashboard (terminal 1)
16 python dashboard/governance_app.py
17 # -> http://localhost:8050
18
19 # 6. Launch storytelling dashboard (terminal 2)
20 python dashboard/storytelling_app.py
21 # -> http://localhost:8051

```

Listing 2: Full startup sequence from a clean clone

6.3. End-to-End Test Results

Verification point	Expected	Result
Bronze DAG run	Success (green)	Pass
Silver DAG run	Success (green)	Pass
Gold DAG run	Success (green)	Pass
governance_*.parquet	Non-empty, 2 rows	Pass
storytelling_*.parquet	Non-empty	Pass
Governance dashboard at :8050	Loads with data	Pass
Storytelling dashboard at :8051	Loads with data	Pass
KPI cards show real values	232 / 73 / 0% / 5.94%	Pass
Sentiment trend covers Apr–May	Date range correct	Pass
Dashboard refresh on new Gold run	Data updates	Pass

7. Challenges and Resolutions

Challenge 1: Negative duplicate rate values in governance Gold

The `duplicate_rate` column showed values between -1 and -0.5 for La Silla Vacía in the Duplicate Rate per Source chart. This was visually confusing and logically invalid.

Resolution: The PySpark deduplication ratio formula had an inverted subtraction ($1 - \text{count}/\text{distinct}$ instead of $1 - \text{distinct}/\text{count}$). The formula was corrected and the Gold DAG re-triggered; all duplicate rates now fall within $[0, 1]$.

Challenge 2: Author null rate appearing in wrong source

The 17.81% null rate for the `author` field was initially attributed to Reddit in the chart, when it belongs to La Silla Vacía (anonymous authors in opinion articles).

Resolution: The Gold aggregation was verified to confirm that Reddit produces zero null authors. The chart legend was corrected and a note was added to the dashboard tooltip explaining that anonymous authorship is expected for La Silla Vacía editorial content.

Challenge 3: Sentiment trend gaps for missing dates

Some calendar dates had no records in either source, producing discontinuous lines in the sentiment trend chart with visible breaks that confused test users.

Resolution: A complete date range was generated with `pd.date_range(start, end, freq='W')` and merged with the aggregated data via left join, filling gaps with `NaN` so Plotly renders a continuous but gapped line (using `connectgaps=False`) rather than a broken axis.

Challenge 4: Keyword label truncation in the top-20 chart

Several high-frequency Spanish political keywords exceeded the default left margin width in the horizontal bar chart, causing label truncation on smaller screens.

Resolution: The Plotly layout margin was set to `dict(l=180)` and the tick font size was reduced to 11px for the keyword axis. The chart height was increased to 500px to prevent bar crowding with 20 items.

8. Conclusions

This workshop brought together four weeks of pipeline development into a working, user-facing product. The two dashboards demonstrate that the Medallion architecture effectively separates concerns: raw data quality is addressed in Bronze and Silver, semantic aggregation happens in Gold, and the presentation layer consumes only clean, pre-aggregated Parquet files without any ad-hoc transformation.

The analysis of Colombian political discourse between April and May 2026 revealed that online opinion is **predominantly neutral**, with negative sentiment outweighing positive on Reddit. The most discussed topics — corruption, electoral campaigns, government performance, and specific candidates such as Cepeda, Paloma, and Abelardo — reflect the heightened political awareness in the months preceding the 2026 presidential election. The spike in publication volume around May 24 suggests that a major political event triggered a significant surge in online discussion.

The contrast between La Silla Vacía (more positive, issue-focused, longer content) and Reddit (more neutral/negative, shorter, more volatile) confirms the value of

combining a curated editorial source with a community-driven platform: together they provide a more complete picture of public opinion than either source alone.

The reproducibility requirement was the most demanding constraint of this workshop and the most valuable: testing the full setup on a clean environment exposed three integration bugs that were invisible during incremental development, including the negative duplicate rate bug and the source label mismatch.

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